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AI-Assisted Development on GitHub

Analyzing AI Contribution Behavior

Thesis BSc Computing Science

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Abstract

Artificial intelligence is becoming increasingly integrated into software development, particularly through tools and agents that can generate code and contribute to collaborative workflows. This thesis investigates how AI generated contributions are integrated and evaluated within collaborative software development on GitHub. Using the AIDEV dataset, the study conducts an analysis of AI authored pull requests, review behavior, AI usage trends in repositories, task type distributions, and differences across AI coding agents.

The results show that AI systems are already actively contributing to software repositories, particularly through pull request creation, and that their usage has increased over time. However, the findings also indicate that review authority and final evaluation remain largely human driven. AI authored pull requests are commonly reviewed through comments and discussion rather than immediate approval, while AI authored reviews are mainly concentrated in feedback oriented behavior. In addition, AI generated contributions are more strongly concentrated in feature implementation, bug fixing, and documentation tasks, whereas human contributions are more broadly distributed, including feature implementation, bug fixing, and documentation like AI generated contributions, but also maintenance and infrastructure related tasks. Differences across AI coding agents are found to be relatively limited.

Overall, the findings suggest that AI is becoming an increasingly visible part of collaborative software development, primarily as a contributor rather than a reviewer or decision maker. This study contributes to a better understanding of the practical role of AI in modern software engineering workflows.

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1 Introduction

Artificial intelligence is becoming increasingly integrated into software development, where it is used to support problem solving, decision making, and a variety of programming related tasks. Rather than serving only as a form of automation, modern AI systems are increasingly positioned as collaborative tools that can improve human capabilities and participate in complex technical workflows [1].

Recent advances in artificial intelligence, particularly in large language models, are changing how software is developed and maintained. Modern AI systems are no longer limited to helping individual developers with isolated tasks. Instead, they are increasingly integrated into collaborative development environments, where they can generate code, suggest modifications, and contribute to ongoing projects [2, 3]. This has introduced a shift in software engineering, where AI systems are not only tools, but active participants in the development process, capable of interacting with human developers and influencing project outcomes [4].

A major part of modern software development takes place on collaborative platforms such as GitHub, where code contributions are proposed, reviewed, and integrated through pull request based workflows [5]. Pull requests serve as the central mechanism for discussing and evaluating code changes before they are merged into a project, making them a key point of interaction between contributors and reviewers [6]. As AI generated contributions increasingly enter these workflows, understanding how they are reviewed, discussed, and accepted becomes essential for evaluating their practical role in collaborative software engineering.

Although AI systems have shown strong capabilities in code generation and task assistance, their practical role in collaborative software development is still not fully understood. In real world development environments, the value of a contribution is not determined solely by whether code can be generated, but also by how that contribution is reviewed, discussed, accepted, and integrated into an existing project. This makes it important to examine not only the productivity potential of AI systems, but also how their contributions are perceived and handled within collaborative workflows.

Existing research has explored AI assisted development from perspectives such as developer productivity, code quality, and large scale contribution patterns [7, 8, 4]. While these studies provide valuable insights into the capabilities and limitations of AI coding systems, less attention has been given to how AI generated contributions are integrated into collaborative development workflows over time. In particular, there remains a need for empirical analysis of how AI authored pull requests are reviewed, how they differ from human contributions, and whether different AI agents exhibit distinct contribution and review patterns in practice.

To address these gaps, this thesis presents an empirical analysis of AI assisted development on GitHub using the AIDEV [4] dataset. The study investigates the extent to which AI generated pull requests contribute to repositories, how

their usage evolves over time, and how these contributions are evaluated within pull request based workflows. In addition, it examines differences between AI generated and human generated review behavior, compares the types of development tasks associated with AI and human contributions, and analyzes whether different AI coding agents exhibit distinct review outcome patterns.
Based on this motivation, the study is guided by the following research questions:
How are AI generated contributions integrated and evaluated within collaborative software development workflows on GitHub?
To answer this main research question in a more structured way, the study is further guided by the following sub research questions:

- SRQ1: To what extent do AI contribute to repositories?
- SRQ2: How has the usage of AI in repositories evolved over time?
- SRQ3: How are AI generated pull requests evaluated by human reviewers?
- SRQ4: How do review behaviors differ between AI generated and human generated reviews?
- SRQ5: How are AI generated contributions evaluated and reviewed within repositories?
- SRQ6: What types of development tasks are AI generated pull requests primarily used for compared to human contributions?
- SRQ7: Do different AI coding agents exhibit different review outcome patterns?

These questions are addressed through an analysis of pull request, review, and task related data from the AIDEV dataset. By examining contribution patterns, review outcomes, temporal trends, and agent specific behavior, this study aims to provide a structured understanding of how AI generated contributions are incorporated into collaborative software development workflows.

2 Background

In this section, the key concepts and technical context necessary for understanding the analyses presented in this thesis are introduced.

2.1 GitHub Workflow

GitHub is a widely used web based source code hosting service for collaborative software development, built on the Git version control system. It enables developers to manage code, track changes, and collaborate through structured workflows [5].

In a typical workflow, developers create separate branches from the main codebase to implement new features or fix issues. Once the changes are ready, they are submitted through a pull request (PR), which serves as a formal proposal to merge the changes into the main branch [6]. A pull request includes not only the code modifications but also descriptive information such as a title and a message explaining the purpose of the changes.
Pull requests act as the central mechanism for collaboration on GitHub. They allow team members to review, discuss, and refine proposed changes before integration. This process ensures that contributions are evaluated for correctness, quality, and alignment with project goals before becoming part of the codebase [6].
This workflow forms the foundation for understanding how both human developers and AI assisted tools contribute to software projects through pull requests.

2.2 PR Review Process

Once a pull request is submitted, it undergoes a review process in which other developers or project maintainers evaluate the proposed changes. This review phase is a central component of the pull based development model [9], enabling collaborative discussion, quality control, and integration of contributions into the main codebase [6].
During the review process, the reviewers examine code changes, provide feedback, and may request modifications before the contribution is accepted. This interaction often involves discussions between contributors and reviewers, where comments are used to clarify implementation details, suggest improvements, or identify potential issues. Previous studies have shown that such discussions are typically concise while still playing an important role in the decision making process [6].
Pull request reviews can result in different outcomes depending on the evaluation. Common outcomes include approval of the changes, requests for modifications, general comments without a final decision, or dismissal of previous reviews. These outcomes reflect how contributions are assessed and whether they are considered suitable for integration into the project.
Previous research has also shown that the success or failure of pull requests is influenced by multiple factors, including project characteristics, developer experience, and the content of review discussions [5]. This highlights the importance of the review process as a decision making stage in collaborative development.
The review process not only ensures code quality but also facilitates knowledge sharing and collaboration among developers. It plays a key role in determining whether a pull request is merged, as well as how long the integration process takes, making it a critical aspect of modern software development workflows.

2.3 Pull Request Task Types

While the GitHub workflow and review process define how contributions are managed and evaluated, understanding the nature of these contributions re-

quires distinguishing between different types of development activities.
Pull requests can represent different types of development activities depending on the nature of the proposed changes. Prior research in software engineering has long emphasized the importance of classifying code changes based on their intent, such as corrective, adaptive, or perfective modifications, in order to better understand development behavior and maintenance activities [10]. In practice, these activities often correspond to changes such as adding new features, fixing bugs, improving performance, refactoring existing code, or performing maintenance tasks. Distinguishing between these task types provides a clearer understanding of development activities and the roles of different contributions within a project.
Classifying pull requests by their purpose enables more detailed analysis of development patterns, such as identifying whether contributors focus on feature development, maintenance, or corrective changes. This categorization is particularly useful when comparing different types of contributors, as it allows for the examination of how various actors participate in different aspects of software development.
In practice, pull request purposes are often described using standardized conventions such as the conventional commit specifications, which defines common categories including `feat`, `fix`, and `chore` [11]. These categories provide a structured way to represent the intent behind code changes.

2.4 AI Coding Agents

Recent advances in artificial intelligence, particularly large language models trained on code, have enabled systems that achieve strong or even human level performance on some programming tasks [12]. This has led to the emergence of AI coding agents that can assist developers or autonomously contribute to software development tasks. These systems are capable of generating code, suggesting modifications, and even creating complete pull requests based on natural language instructions or contextual information from the codebase [4].
Examples of AI coding agents relevant to this study include GitHub Copilot [13], Devin [14], Cursor [15], OpenAI Codex [16], and Claude Code [17], which are also the five agents represented in the AIDEV dataset [4]. These agents leverage large language models trained on large scale code and natural language data to understand programming context and produce relevant code suggestions [7].
Unlike traditional development tools, AI coding agents can act as active contributors within collaborative workflows. Recent work in software engineering argues that AI systems are increasingly evolving from assistive coding tools toward more collaborative roles in development environments, where they may support or participate in broader software engineering activities [18]. This introduces a new paradigm in software engineering, where AI systems can function as collaborators rather than passive tools.
The integration of AI agents into development workflows raises important questions about how their contributions are evaluated, how they differ from human generated contributions, and how they influence collaboration dynamics.

Understanding these aspects is essential for analyzing the role of AI assisted development in modern software engineering.

3 Related Work

The rapid advancement of artificial intelligence, particularly large language models, has led to growing interest in their use for software engineering tasks. Recent literature shows that AI techniques are increasingly being integrated across multiple phases of the software engineering lifecycle, including planning, requirements engineering, development, testing, deployment, and maintenance [19]. Rather than being limited to isolated assistive roles, these systems are increasingly integrated into software development workflows through functionalities such as code generation, testing support, documentation, and code review [20].

Some studies focus on the productivity impact of AI coding assistants. Peng et al. conducted a controlled experiment to evaluate the impact of GitHub Copilot on programming tasks, showing that developers using Copilot completed tasks up to 55.8% faster than those without AI assistance [7]. These results point to the potential of AI tools to accelerate software development. However, controlled experiments may not fully reflect real world collaborative workflows, where factors such as code review, integration, and team dynamics play an important role.

Recent work has also examined how AI coding assistants affect collaborative software development in real world open source environments. Song et al. investigated the impact of GitHub Copilot on collaborative open source software development and found that Copilot usage was associated with increased project level code contributions and greater developer participation, but also with longer coordination time for integrating contributions [21]. These results suggest that while AI tools may improve productivity, they can also introduce additional coordination overhead in collaborative workflows. This makes it important to study AI not only as a productivity tool for individual developers, but also as a factor that influences interaction and review dynamics within software teams.

Although AI coding assistants have shown clear productivity benefits, several studies have raised concerns about the quality and reliability of AI generated code. Yetiştiren et al. evaluated GitHub Copilot, Amazon CodeWhisperer [22], and ChatGPT [23] in terms of code validity, correctness, security, reliability, and maintainability, and found that fully correct code generation remained inconsistent across tools, with correctness rates ranging from 31.1% to 65.2% depending on the system [8]. Their findings suggest that while AI tools can accelerate development, the generated outputs still require human review and oversight to ensure correctness and maintainability.

Recent work has also started to examine the role of AI agents in real world software engineering environments. Li et al. introduced the AIDEV dataset, a large scale collection of pull requests authored by autonomous coding agents across thousands of GitHub repositories [4]. This study shows that AI agents

are no longer limited to assistive roles, but actively participate in development workflows by generating pull requests and interacting with human collaborators. However, although these agents can increase contribution rates, AI generated pull requests were found to be less likely to be accepted than human authored contributions, suggesting a gap between productivity and practical usefulness in real world settings.

Although these studies provide useful insights into the capabilities and impact of AI assisted development, they primarily focus on high level metrics such as task completion time, acceptance rates, and overall contribution patterns. Less attention has been given to understanding how AI assisted development evolves over time, how AI generated contributions are reviewed in practice, and how different AI agents behave within collaborative environments.

To address these gaps, this study presents a detailed analysis of AI assisted development on GitHub using the AIDEV dataset. Specifically, temporal trends in AI usage, review dynamics of AI generated contributions, differences between human and AI reviews, and behavioral variations across different AI agents are investigated. This work aims to provide a deeper understanding of how AI is integrated into modern software development workflows by investigating these aspects.

4 Methodology

4.1 The Dataset

The dataset used in this research, called AIDEV [4], contains nearly one million pull requests across more than 116,000 GitHub repositories. It captures the emergence of AI coding agents as collaborators in real world open source software development. These pull requests are authored by over 72,000 human developers as well as five AI coding agents: OpenAI Codex, Devin, GitHub Copilot, Cursor, and Claude Code. Each pull request is associated with its corresponding repository and author, along with additional metadata such as timestamps, review data, and textual content.

AI authored pull requests are identified based on the `agent` label provided in the dataset, which specifies the AI system responsible for generating the contribution.

The dataset is organized into normalized tables that can be joined through consistent identifiers, enabling integration of pull requests, reviews, comments, and repository metadata. An overview of the dataset schema of the tables that are being used in this research can be seen below in Figure 1.

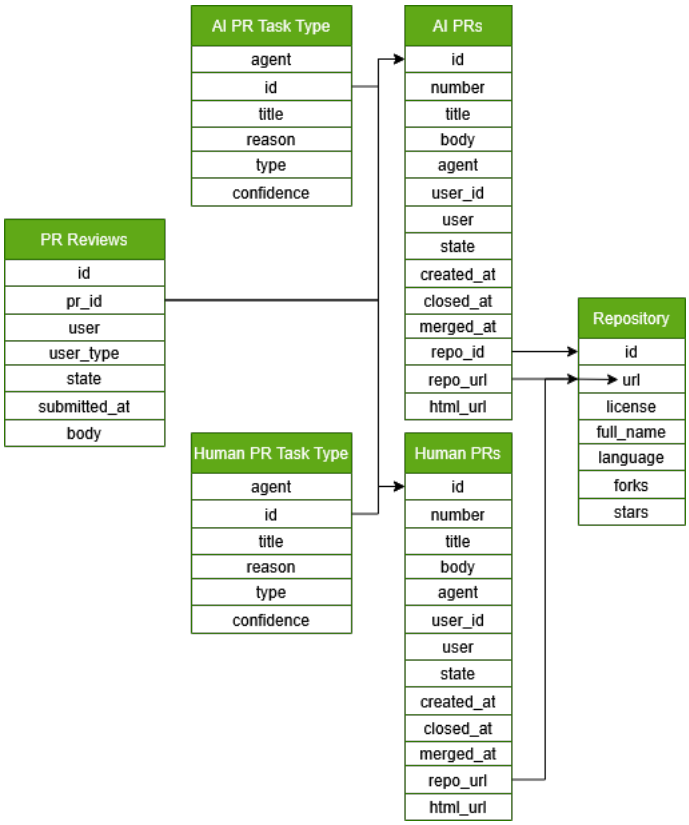


Figure 1: Dataset Schema

For the purposes of this study, multiple components of the dataset were utilized, including AI authored pull requests from repositories with more than 100 stars and human authored pull requests sampled from the same repositories, restricted to those with more than 500 stars. In addition, repository data, task type annotations, and pull request reviews were utilized.

Task type annotations are automatically generated using a language model based on conventional commit categories, providing a standardized classification of pull request purposes [4]. Personal identifiable information has been removed from the dataset to address privacy concerns.

The structure of the datasets used in this research is presented in the following tables, along with their key attributes and descriptions.

Attribute	Description
id	Identifier for the pull request
number	Pull request number within the repository
title	Title of the pull request
body	Markdown body submitted with the pull request
agent	The AI Agent used for the pull request
user_id	Identifier of the pull request author
user	Username of the pull request author
state	Current status of the pull request (open, closed)
created_at	Timestamp when the pull request was created
closed_at	Timestamp when the pull request was closed (if applicable)
merged_at	Timestamp when the pull request was merged (if applicable)
repo_id	Identifier of the repository the PR belongs to
repo_url	API URL of the repository
html_url	Web URL for viewing the pull request on GitHub

Table 1: Description of AI Pull Requests Dataset

Attribute	Description
id	Identifier for the pull request
number	Pull request number within the repository
title	Title of the pull request
body	Markdown body submitted with the pull request
agent	Human
user_id	Identifier of the pull request author
user	Username of the pull request author
state	Current status of the pull request (open, closed)
created_at	Timestamp when the pull request was created
closed_at	Timestamp when the pull request was closed (if applicable)
merged_at	Timestamp when the pull request was merged (if applicable)
repo_url	API URL of the repository
html_url	Web URL for viewing the pull request on GitHub

Table 2: Description of Human Pull Requests Dataset

Attribute	Description
id	Identifier for the repository
url	API URL of the repository
license	SPDX identifier for the repository license (if applicable)
full_name	Owner name of the repository
language	Primary coding language used in the repository reported by GitHub's linguist
forks	Count of forks that the repository has
stars	Count of starsgazer that the repository has

Table 3: Description of Repository Dataset

Attribute	Description
agent	The AI Agent used for the pull request
id	Identifier for the pull request
title	Title of the pull request
reason	Reason for the pull request
type	Pull request type (e.g. feat, chore, fix, other etc.)
confidence	Confidence rate of the pull request out of 10

Table 4: Description of AI Pull Requests Task Type Dataset

Attribute	Description
agent	Human
id	Identifier for the pull request
title	Title of the pull request
reason	Reason for the pull request
type	Pull request type (e.g. feat, chore, fix, other etc.)
confidence	null

Table 5: Description of Human Pull Requests Task Type Dataset

Attribute	Description
id	Identifier for the review
pr_id	Identifier for the pull request associated with the review
user	Username of the review author
user_type	User type of the review author (Bot or User)
state	State(type) of the review (e.g. Commented, Approved etc.)
submitted_at	Timestamp when the review was made
body	Review markdown body

Table 6: Description of Pull Request Reviews Dataset

4.2 Data Preparation and Tools

The data processing and analysis were conducted primarily using Python on Google Colab, which is a cloud based environment. The datasets were obtained from Hugging Face [24] and loaded using the `pandas` [25] library, which was also used for data cleaning, filtering, merging, and aggregation operations throughout the study.
The data was stored in parquet format, allowing efficient handling of large scale datasets. During preprocessing, relevant attributes were selected, and transformations such as grouping, time based aggregation, and normalization were applied to prepare the data for analysis.
For visualization, the `matplotlib` [26] library was used to generate plots that illustrate trends and distributions across different analyses. These visualizations were used to support the interpretation of results presented in the subsequent sections.
The specific methods used for processing the data and generating statistical results are described in detail in the following sections.

4.3 Data Processing and Analysis

This section describes the procedures used to process the dataset and perform the analyses required to answer the research questions. The processing steps applied to the dataset have been described in the following sections.

4.3.1 Pull Request Analysis

For this analysis, AI and Human pull requests datasets were used. Each pull request dataset was grouped by repository using the `repo_url` attribute to find how many bot and human pull requests each repository has. This resulted in two separate tables that contain repositories with their repository URL and their AI or Human pull request count.

Repository URL	**AI PR Count**
https://api.github.com/01	2
https://api.github.com/02	2
https://api.github.com/03	18
...	...

Table 7: Example of AI Pull Request Count per Repository

Repository URL	Human PR Count
https://api.github.com/01	4
https://api.github.com/04	5
https://api.github.com/05	1
...	...

Table 8: Example of Human Pull Request Count per Repository

Next, these tables were merged on `repo_url` attribute, resulting in a combined table that contains the repository url, AI pull request count, and human pull request count.
To ensure comparability between AI generated and human generated pull request datasets, filtering the combined PR datasets for repositories that have 500 or more stars was necessary, since without applying a star based filter, the distribution of repositories would be naturally biased towards AI heavy repository due to the differences in the original sampling thresholds. To do this, all attributes other than `url` and `stars` were removed from the repository dataset, and the resulting table was joined with the merged pull request counts table on the `repo_url` attribute.

URL	AI PR	Human PR	Stars
https://api.github.com/01	2	4	5609
https://api.github.com/06	26	0	514
https://api.github.com/07	10	11	1737
...	...	...	...

Table 9: Example of Pull Request Counts with Stars per Repository

Two additional attributes were added during the processing.
The first, the total pull request count, represents the total number of pull requests submitted to a repository. This makes it possible to identify and filter out repositories that have low activity. For this analysis, repositories with less than 5 pull requests were filtered out.
The second, the percentage of AI, represents the proportion of AI generated pull requests relative to the total number of pull requests for that repository. This metric is used to compare repositories based on their level of AI usage.

URL	AI PR	Human PR	Stars	Total PR	AI %
https://api.github.com/01	2	4	5609	6	33.3
https://api.github.com/06	26	0	514	26	100.0
https://api.github.com/07	10	11	1737	5	47.6
...	...	...	...	...	...

Table 10: Example of Pull Request Counts and AI Percentage per Repository

4.3.2 AI Rate of Pull Request Over Time

For this analysis, AI and Human pull requests datasets were used. First, the `created_at` attribute was converted into a Year-Month-Day format. The datasets were then grouped by month to analyze the number pull requests authored by AI and human per month.

Year-Month	AI PR Count
2024-12	65
2025-01	412
2025-02	539
...	...

Table 11: Example of AI Pull Request Count per Month

Year-Month	Human PR Count
2025-01	940
2025-02	1002
2025-03	1133
...	...

Table 12: Example of Human Pull Request Count per Month

Next, these tables were merged on `month` attribute. This produced a table that contains the attributes month, AI pull request count, and human pull request count. Then a new attribute AI Percentage was added to represent the AI pull request rate of the month.

Year-Month	AI PR Count	Human PR Count	AI %
2024-12	65	0	100.0
2025-01	412	940	30.47
2025-02	539	1002	34.97
...	...	...	...

Table 13: Example of AI Percentage per Month

4.3.3 Review Analysis

For this analysis, pull request reviews dataset and both AI and human pull request datasets were used. The pull request datasets were needed to make a repository based analysis. The pull request reviews dataset does not have a repository related attribute, so the `pr_id` attribute was to merge it to pull request datasets and use the `repo_url` attribute from a combined pull request dataset to make a repository based analysis.
First, human and AI pull request datasets were combined . After that, attributes

that are not needed were dropped, so only `id` and `repo_url` from the combined pull requests table remained and merged with the pull request reviews dataset on the `pr_id` attribute. While merging, attributes that are not needed were dropped once more so only `user_type`, `state`, `stars` and `repo_url` attributes remained.

User Type	**State**	**Stars**	**Repo URL**
Bot	Commented	267	https://api.github.com/10
Human	Approved	124	https://api.github.com/11
Bot	Dismissed	153	https://api.github.com/12
...	...	...	...

Table 14: Example of Reviews Dataset with Repository URL

The dataset was then separated into two tables based on the `user_type` attribute. After preprocessing, these tables were merged again for further analysis. This attribute can be either `User` which indicates that the review is done by a human, or `Bot` which indicates that the review is done by an AI bot. This resulted in two tables that represent AI and human authored reviews.

User Type	**State**	**Stars**	**Repo URL**
Bot	Commented	267	https://api.github.com/10
Bot	Dismissed	153	https://api.github.com/12
Bot	Changes Requested	543	https://api.github.com/15
...	...	...	

Table 15: Example of AI Reviews Dataset with Repository URL

User Type	**State**	**Stars**	**Repo URL**
Human	Approved	124	https://api.github.com/11
Human	Approved	623	https://api.github.com/13
Human	Commented	812	https://api.github.com/14
...	...	...	

Table 16: Example of Human Reviews Dataset with Repository URL

These two tables were used to process the data further in two different ways.

4.3.3.1 Review Counts per Repository

Each review table was grouped by repository using the `repo_url` attribute to calculate the review number of AI authored and human authored reviews per repository. The resulting tables were then merged on `repo_url`, and two new attributes called total review count and percentage of AI were added. Total review count is the AI review count and the human review count added together,

and AI percentage is AI review count divided by total review count. The data was then filtered to retain only repositories with more than 5 reviews and at least 500 stars. This ensures comparability by identifying and filtering out repositories with low activity.

URL	AI Rev.	Human Rev.	Stars	Total Rev.	AI %
https://api.github.com/17	43	11	4722	54	79.6
https://api.github.com/18	5	4	919	9	55.5
https://api.github.com/19	3	12	826	15	20.0
...	...	...	...	...	...

Table 17: Example of Review Counts and AI Percentage per Repository

4.3.3.2 Review Counts per Type

Each review table was grouped by their issue state using the `state` attribute to find the AI and human review count per issue state. Then these two tables were added together, having AI review count and human review count per state together in a table. In this case, filtering out reviews from repositories with fewer than 500 stars or low activity levels was not considered necessary, as the analysis does not compare repositories directly. Instead, the objective is to examine differences in review state distributions between AI authored and human authored reviews. Since the available review data from repositories with more than 100 stars was considered sufficient for this purpose, no additional filtering was applied.

State	AI Review Count	Human Review Count
Approved	76	6422
Changes Requested	98	1506
Commented	11741	8690
Dismissed	64	277

Table 18: Example of Review Counts per Issue State

4.3.4 Pull Request Issue State

The goal of this analysis is to merge pull request dataset to pull request review dataset, and find the issue state of each pull request. Unfortunately, for human pull requests, review data doesn't exist. So this processing and analysis of data will only be for AI authored pull requests.
First, the reviews with `pr_id` that does not exist within the `id`'s of the AI pull request dataset were filtered out. This resulted in a dataset of reviews that are only associated with AI authored pull requests. Attributes other than `state` were also removed, since that is the only attribute needed for this analysis. Since only the review states assigned to AI authored pull requests were relevant for this analysis, the `state` attribute was retained as the only necessary variable.

The dataset was then grouped by `state` to calculate how many reviews of each type were assigned to AI authored pull requests.

State	Count
Approved	6499
Changes Requested	1604
Commented	20431
Dismissed	341

Table 19: Example of Review Counts per Issue State for AI PRs

4.3.5 Task Type

For this analysis, the AI pull request task type and human pull request task type datasets were used. A task type for each pull request is assigned using LLM classification on pull request titles and commit messages. Each dataset was grouped by `type` to see the task distribution for both AI and human pull requests.

Task Type	AI PR Count
build	627
chore	896
ci	411
docs	3887
...	...

Table 20: Example of Task Type Counts for AI PRs

Task Type	Human PR Count
build	614
chore	847
ci	100
docs	519
...	...

Table 21: Example of Task Type Counts for Human PRs

4.3.6 Approvement Rate of AI PRs

For this analysis, AI pull requests and pull request reviews datasets were used. These tables were merged on the pull request ID, and the `created_at` attribute from the AI PRs dataset is transformed to a Year-Month-Day format.

PR ID	Year-Month	State
111	2025-07	Commented
123	2025-07	Approved
456	2025-07	Commented
...	...	...

Table 22: Example of AI PRs with State and Time

Next, the dataset was grouped first by month, and then by state. This made it possible to calculate the state count per each month for AI pull requests. An additional percentage attribute was also added to represent the proportion of each review state within a given month.

Year-Month	State	Count	Percentage
2024-12	Approved	20	31.74
2024-12	Changes Requested	3	4.76
2024-12	Commented	40	63.49
...	...	...	...

Table 23: Example of AI PRs with State and Time

4.3.7 Review Types per Agent

The AI Dev dataset contains agentic pull request data authored by five different agents. AI pull requests dataset contains an `agent` attribute that indicates the author of the pull requests.
The AI pull requests and pull request reviews were merged and grouped by `agent` and `state` attributes respectively to see the review state of PRs done by each agent.

Agent	State	Count
Claude_Code	Approved	103
Claude_Code	Changes Requested	15
...	...	...
Copilot	Approved	3257
Copilot	Changes Requested	1092
...	...	...

Table 24: Example of Review States per Agent

4.3.8 Confidence Rate per Agent

For this analysis, the pull request task type dataset was used, focusing on the `agent` and `confidence_rate` attributes. The `agent` attribute represents the AI system responsible for generating the pull request, while the `confidence_rate` attribute provides a score between 1 and 10 indicating how confidently the task

type of the pull request was classified by a language model.
First, the dataset was filtered to include only entries with valid agent and confidence values. The data was then grouped by agent to compute two metrics, the average confidence and its standard deviation for each agent. The average confidence reflects how clearly the tasks generated by each agent can be classified, while the standard deviation indicates the variability of this confidence across pull requests.

Agent	**Avg. Conf. Rate**	**STD Conf. Rate**
Claude Code	9.710	0.491
Devin	9.620	0.549
Copilot	9.478	0.551
Cursor	9.284	0.735
OpenAI Codex	9.099	0.596

Table 25: AI Confidence Rate per Agent

5 Results

This section presents the results of the analyses conducted on the dataset. The findings are organized according to the research questions and cover multiple aspects of AI assisted development, including repository level contribution patterns, temporal trends, review dynamics, task types, and differences across AI agents.

5.1 AI Usage Across Repositories

AI Rate	**Count**
Mixed	286
Tends AI	106
Fully AI	94
Mostly AI	29
Tends Human	1
Mostly Human	0
Fully Human	0

Table 26: AI Rate of Repositories Based on PR Authors

AI Rate	Count
Mixed	99
Tends Human	98
Fully Human	81
Mostly Human	61
Tends AI	52
Mostly AI	34
Fully AI	21

Table 27: AI Rate of Repositories Based on Review Authors

To better understand how strongly repositories rely on AI generated contributions, each repository was categorized according to their AI authored pull requests and reviews percentage. The computation of AI percentages for pull requests and reviews were described earlier in sections 4.3.1 and 4.3.3.1.
After the AI percentage for each repository was obtained, these values were mapped into categories for clearer comparison. The categorization scheme was defined as follows:

- 0% → Fully Human
- 1–20% → Mostly Human
- 21–40% → Tends Human
- 41–60% → Mixed
- 61–80% → Tends AI
- 81–99% → Mostly AI
- 100% → Fully AI

Tables 26 and 27 present the resulting distributions when this categorization is applied to pull request authorship and review authorship, respectively.
The pull request data show a clear skew toward AI driven contribution, whereas review activity remains more human centered.

SRQ1: To what extent do AI contribute to repositories?

The results indicate that AI generated pull requests are widely present across the analyzed repositories. Most repositories fall into the Mixed or AI leaning categories when considering pull request authorship, indicating that AI assisted contributions frequently appear alongside human contributions. In contrast, review activity remains more human centered. The majority of repositories fall into human leaning categories when review authorship is considered, suggesting that while AI tools are increasingly used to generate pull requests, the review and validation process is still primarily carried out by human developers.

5.2 Monthly Percentage of AI Authored Pull Requests

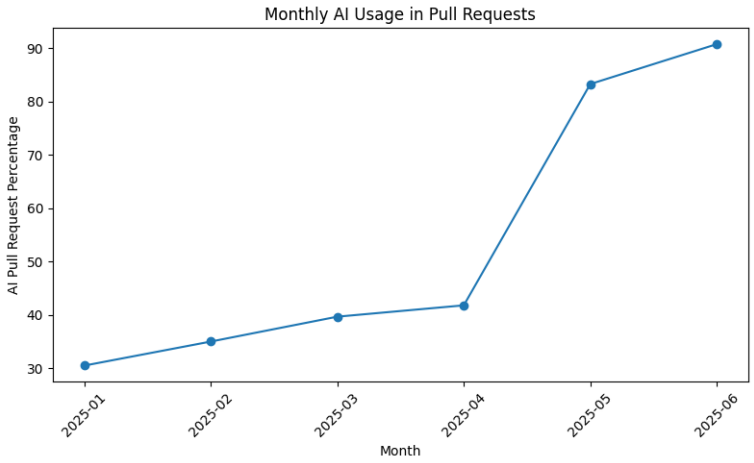


Figure 2: Monthly Percentage of AI Authored Pull Requests

In Section 4.3.2, the monthly percentage of AI authored pull requests was calculated and a slice of it was presented in tabular form. To identify patterns and observe changes over time, the same data was visualized on a graph.
As shown in Figure 2, the proportion of AI authored pull requests increases steadily from January to April 2025, followed by a sharp increase beginning in May 2025. While this trend suggests a rapid increase of AI assisted development, the sudden rise raises the possibility that the dataset itself may have expanded in later months, introducing additional repositories with high AI activity.
To account for this, an additional analysis was performed by filtering the dataset to include only repositories that have entries in all observed months. This ensures that the trend reflects consistent repository activity rather than the introduction of new repositories.

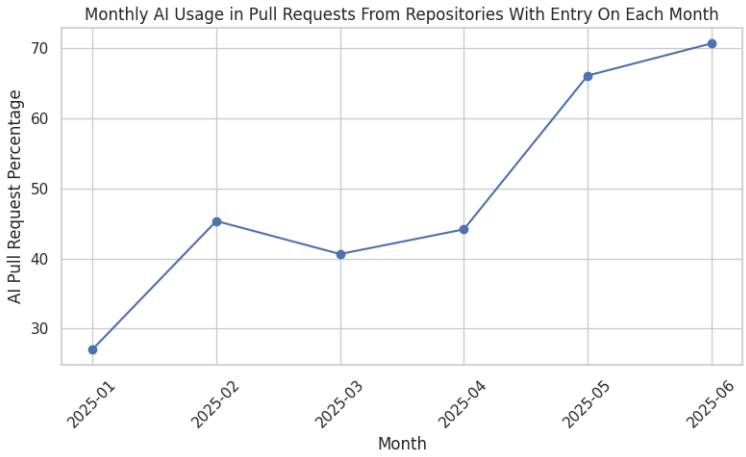


Figure 3: Monthly AI Usage in Pull Requests for Repositories Active Across All Months

Figure 3 presents the results of this filtered analysis. Compared to the initial graph, the increase in AI authored pull requests remains evident but appears more gradual and consistent. The sharp jump observed in the earlier figure is reduced, indicating that part of the increase was influenced by newly introduced repositories. Nevertheless, the overall trend is still showing that AI assisted development usage increases over time even when controlling for repository activity.

SRQ2: How has the usage of AI in repositories evolved over time?

The results show that the usage of AI generated pull requests increases over time. The initial analysis suggests a sharp rise in AI authorship in later months, however, further examination reveals that part of this increase is influenced by the introduction of new repositories.
When considering only repositories that are active across all observed months, the trend remains upward but becomes more gradual and consistent. This indicates that AI assisted development is progressively adopted within repositories over time, rather than increasing solely due to dataset expansion.

5.3 Review Outcomes of AI Authored Pull Requests Over Time

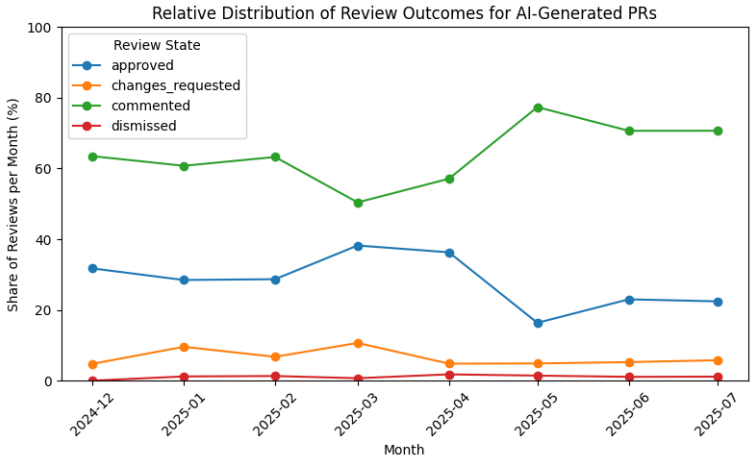


Figure 4: Distribution of Review Outcomes for AI Authored Pull Requests Over Time

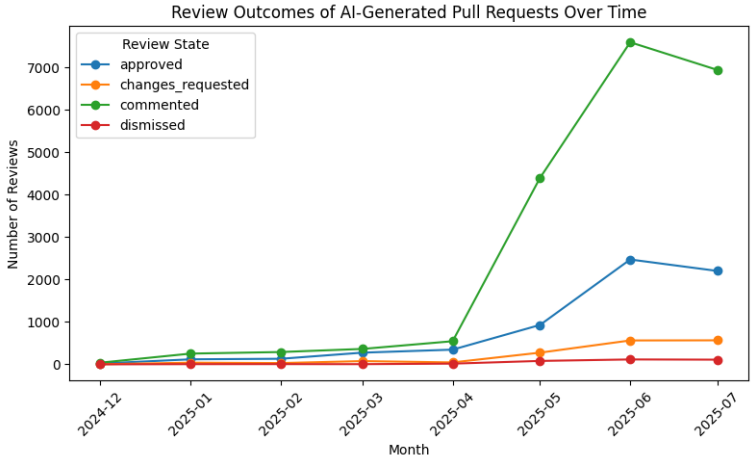


Figure 5: Number of Review Outcomes for AI Authored Pull Requests Over Time

In Section 4.3.6, the reviews of AI authored pull requests were examined and a part of the results was presented in tabular form. To better understand whether the effectiveness of AI contributions changes over time, the results were visualized in Figures 4 and 5. The first figure shows the distribution of review states per month, while the second presents the total number of reviews in each category.

As shown in Figure 4, the distribution of review outcomes remains stable across the observed period. The share of approved, changes_requested, commented, and dismissed reviews fluctuates only slightly, without indicating a strong trend in any review type. This suggests that AI generated pull requests are evaluated with consistent standards over time.
In Figure 5, a substantial increase in the number of reviews across all categories can be seen. This reflects the growth in AI assisted contributions, leading to a higher volume of AI authored pull requests entering the review process.
Together, these findings indicate that while the quantity of AI contributions increases, their evaluation pattern remains largely unchanged. This stability suggests that the evaluation patterns of AI generated pull requests remain consistent over time.

SRQ3: How are AI generated pull requests evaluated by human reviewers?

The results show that the evaluation patterns of AI authored pull requests remain relatively stable over time. The proportional distribution of review outcomes does not change significantly across the observed months, with most reviews classified as comments followed by approvals, while change requests and dismissals remain comparatively rare.
At the same time, the total number of reviews increases substantially during the later months, indicating that AI generated pull requests become more frequent. Despite this growth, the relative distribution of review outcomes remains consistent, suggesting that the perceived quality and review standards applied to AI generated contributions remain stable even as their usage increases.

5.4 Review Type Results

State	AI % of States	AI's Distribution	Human's Distribution
Approved	1.17	0.63	38.01
Changes Requested	6.11	0.82	8.91
Commented	57.47	98.01	51.44
Dismissed	18.77	0.53	1.64

Table 28: Review State Distribution of AI and Human Authors

To further investigate how AI agents participate in the code review process, the distribution of review states authored by AI and human contributors were analyzed . The procedure used to extract and aggregate review data was described previously in Section 4.3.3.2, where reviews were grouped by their `state` attribute and counted separately for AI and human authors.
Table 28 presents three complementary perspectives for each review state.

- **AI % of States** indicates the proportion of reviews in a given state that were authored by AI bots. For example, only 1.17% of all approved reviews were written by AI.
- **AI's Distribution** represents how AI authored reviews are distributed across states. This shows what AI systems primarily do when they participate in reviews. For example, only 0.63%, 0.82%, and 0.53% of AI authored reviews fall into the approved, changes requested, and dismissed categories, respectively.
- **Human's Distribution** provides the same distribution for human authored reviews. For example, 38.01% and 51.44% of human authored reviews have the state approved and commented, indicating that these actions constitute a substantial portion of human review activity.

The results show a strong behavioral concentration: nearly all AI authored reviews fall into the commented category (98.01%), while approval and change request actions are mostly performed by human reviewers. In contrast, human review activity is more evenly distributed and includes a substantial proportion of approvals (38.01%) and change requests (8.91%).
This suggests that, within the dataset, AI tools are used primarily for supportive or observational feedback rather than for decisive review actions such as approving or rejecting changes. Consequently, AI appears to function more as an assistive participant in the review workflow, whereas humans retain the role of evaluators responsible for merge decisions.

SRQ4: How do review behaviors differ between AI generated and human generated reviews?

The results indicate a clear difference between AI and human reviewing behavior. Vast majority of AI authored reviews are classified as commented, accounting for almost all AI review activity. In contrast, human reviewers produce a more balanced distribution of review states, including approvals and change requests.
This suggests that AI systems are primarily used to provide feedback or discussion rather than to make final review decisions. Human contributors therefore remain responsible for the decisive evaluation steps in the review process.

5.5 Review Outcomes of AI Authored Pull Requests

State	Count	Percentage
Approved	6499	22.51
Changes Requested	1604	5.55
Commented	20431	70.76
Dismissed	341	1.18

Table 29: Reviews of AI Authored PRs

Table 29 presents the review outcomes assigned to pull requests authored by AI agents. The majority of these pull requests (70.76%) received reviews classified as commented, while only 22.51% were directly approved. A small proportion resulted in requests for changes (5.55%) and very few were dismissed (1.18%). These results suggest that AI generated pull requests are typically not treated as finalized contributions, but rather as drafts that invite discussion and refinement. Reviewers appear to engage with AI generated code by providing explanatory or corrective feedback before making an approval decision. The low number of dismissals indicates that AI contributions are rarely rejected outright, which implies that they typically meet baseline project requirements or are on the right track.

SRQ5: How are AI generated contributions evaluated and reviewed within repositories?

The results show that AI authored pull requests are most frequently reviewed through comments rather than immediate approval or rejection. The majority of reviews on AI generated pull requests are classified commented, while approvals represent a smaller but still significant portion of outcomes. Requests for changes and dismissals occur comparatively infrequently.
This pattern suggests that AI generated pull requests are typically treated as draft contributions that invite discussion and refinement before final acceptance. Reviewers appear to engage with these contributions by providing feedback and clarification rather than making immediate approval decisions.

5.6 Task Type Results

Task Type	AI's %	Human's %
Build	1.87	9.28
Chore	2.67	12.80
CI	1.22	1.51
Docs	11.57	7.84
Feature	43.01	29.40
Fix	24.13	26.87
Performance	1.01	1.33
Refactor	6.81	5.58
Revert	0.05	0.00
Style	0.56	0.89
Other	0.09	1.74

Table 30: Task Type Distribution for AI and Human Contributers

The results of the task type analysis are presented in Table 30. The methodology used to derive these distributions was described earlier in Section 4.3.5, where pull requests were categorized according to their labeled development task (e.g., Feature, Fix, Refactor).

The table shows the share of each task type among AI and human authored pull requests. Each percentage represents how frequently a given task occurs within the total contributions of that author group. For example, 43.01% of AI generated pull requests are related to implementing new features, where 29.40% of human authored pull requests fall into the same category.

Both AI and human contributors concentrate heavily on feature development and bug fixing, indicating that AI assisted development is being applied to core implementation tasks rather than only maintenance activities. Smaller categories such as Style, Revert, or Performance remain relatively rare for both groups, suggesting these tasks are either less common overall or less frequently delegated to AI tools.

Despite the overall similarities, there are some notable differences. AI authored pull requests show a stronger concentration in feature related tasks (43.01%) compared to humans (29.40%), suggesting that AI tools are frequently used to generate new functionality. Human contributors exhibit higher involvement in maintenance oriented categories such as chore (12.80% vs. 2.67%) and build tasks (9.28% vs. 1.87%), which often require contextual project knowledge, dependency management, or environment configuration.

Overall, these differences suggest that AI is primarily leveraged for forward development activities, while humans remain more engaged in project maintenance and integration tasks.

SRQ6: What types of development tasks are AI generated pull requests primarily used for compared to human contributions?

The results show that AI generated pull requests are most frequently associated with feature implementation and bug fixes, which together constitute the majority of AI contributions. Feature related tasks represent the largest category for AI authored pull requests, indicating that AI tools are commonly used to generate or scaffold new functionality.
In contrast, human authored pull requests show a more balanced distribution across task types and appear more frequently in maintenance oriented categories such as chore and build. These findings suggest that while AI systems are actively used for core development tasks, human developers remain more involved in project maintenance, integration, and environment related changes.

5.7 Agent Results

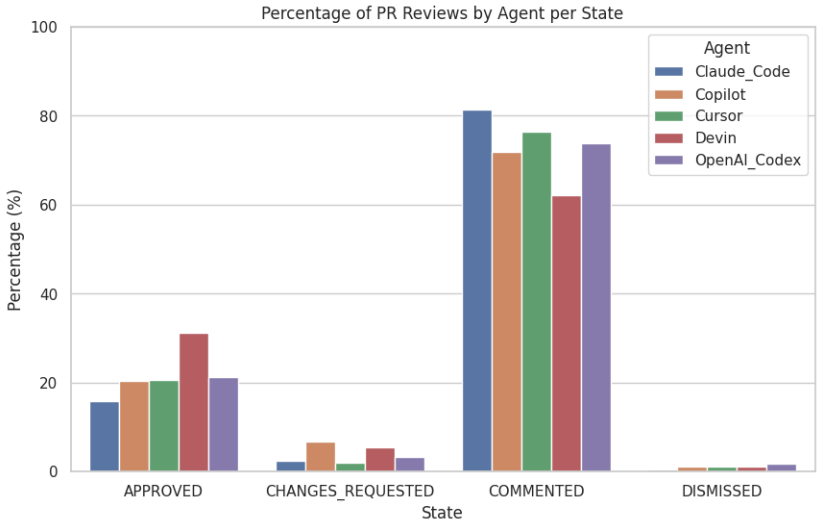


Figure 6: Distribution of Review Outcomes for Pull Requests by each AI Agent

To further investigate differences within AI assisted development, an agent level analysis was conducted by distinguishing between the five AI contributors present in the dataset. These agents are, Claude Code, Copilot, Cursor, Devin, and OpenAI Codex. This analysis examined whether different agents receive different types of review outcomes. In Section 4.3.7, steps taken for this analysis are shown and explained.
Figure 6 show a consistent pattern for all agents:

- Majority of the reviews are comments.
- Although approved reviews has the second largest share, it's substantially lower compared to comments.
- Change request occur very infrequently.
- Pull requests are rarely dismissed.

Although the results are broadly similar across all AI agent, there are minor differences.
For example, Devin shows the highest approval share compared to other AI agents. It also gets less comments from the reviewers. Devin, has the second most change request rate for its pull requests, slightly behind Copilot. However, the overall values remain close to one another, making it difficult to distinguish a single agent as significantly more successful or less successful than the others.

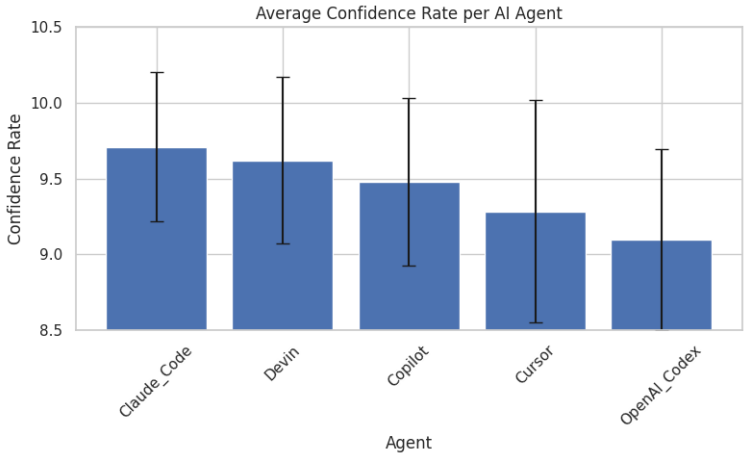


Figure 7: Average and STD of Confidence Rate by each AI Agent

To complement the analysis of review outcomes, the average confidence rate of task classification for each agent was also examined, where the steps taken for this are shown in Section 4.3.8. As shown in Figure 7, all agents exhibit high confidence scores, with average values above 9.0. While small differences are visible, such as Claude Code having the highest average confidence and OpenAI Codex the lowest, these variations remain limited.
Additionally, the standard deviation values are relatively small across all agents, indicating that the confidence scores are consistent and do not vary significantly between pull requests.
Together with the review outcome analysis, these results reinforce the conclusion that differences between AI agents are minor, both in terms of how their contributions are evaluated and how clearly their tasks can be categorized.

SRQ7: Do different AI coding agents exhibit different review outcome patterns?

The results indicate that the review outcome distributions of the analyzed AI agents are largely similar. Across all agents, the majority of reviews fall into the commented category, followed by approved, while changes requested and dismissed remain comparatively rare.
Although minor differences exist, for example, Devin shows a slightly higher approval share and Copilot a slightly higher rate of change requests, these variations are relatively small. Overall, reviewers appear to interact with pull requests generated by different AI agents in broadly similar ways.
Additionally, the confidence rate analysis shows that all agents produce tasks with consistently high confidence scores, with only small differences in average values and variability. This suggests that the outputs of different AI agents are similarly clear and well defined.
Taken together, these findings indicate that differences between AI agents are limited, both in terms of how their contributions are evaluated and how clearly their tasks can be categorized.

6 Discussion

The results of this study shows a clearer view on how AI generated contributions are currently used in collaborative software development workflows on GitHub. Across the analyses, a consistent pattern can be observed. AI systems are widely used to generate pull requests and propose changes, but they play a much smaller role in reviewing contributions or deciding whether those changes should be accepted. This suggests that AI is mainly used as a tool for producing code, while human developers remain responsible for evaluating and integrating those contributions. This section discusses the implications of these findings in relation to collaborative development, review practices, and the evolving role of AI in software engineering.

6.1 AI as a Contributor Rather Than a Reviewer

One of the clearest findings of this study is that AI is more strongly represented in contribution activity than in review authority. As shown in Tables 26, AI generated pull requests constitute a substantial portion of contributions within the analyzed repositories, indicating that AI systems are actively participating in development workflows. In contrast, the result Table 27 suggest that the evaluation and interpretation of these contributions still remain largely human driven. Although the AI participation in review activity is present, Table 28 demonstrates that AI authored reviews are primarily concentrated in the `commented` category, accounting for the majority of AI review activity. In contrast, approvals and change requests are primarily performed by human reviewers. This indicates that, although AI systems participate in review discussions, decisive

evaluation actions remain largely human driven.
This suggests that AI is currently integrated into collaborative software development primarily as a contributor rather than as a reviewer or decision maker. In practice, AI appears to function more as a generator of proposed changes, while human developers retain responsibility for validating, interpreting, and deciding whether those contributions should be accepted. One possible explanation is that review decisions require contextual understanding of project standards, long term design considerations, and implicit team conventions, which current AI systems may not fully capture. Overall, this suggests that AI is currently used to support development work, but not to replace the human judgment that is still needed in collaborative software projects.

6.2 Increasing AI Usage Over Time

An important finding of this study is the clear increase in AI generated pull requests over time. As shown in Figure 2, the monthly share of AI generated contributions rises substantially across the observed period, suggesting that AI tools are becoming more widely used within GitHub based development workflows.
However, this trend should be interpreted with caution. As discussed in Section 5.2, the raw increase in AI usage may partly reflect changes in the composition of the dataset over time, rather than only a true increase in AI usage within the same set of repositories. In particular, the dataset documentation [4] indicate that some AI agents, especially OpenAI Codex, become much more common in later months, which may inflate the appearance of continuous growth.
To address this, an additional controlled analysis was performed using only repositories that were active across all observed months. As shown in Figure 3, the upward trend remains visible even after this filtering step, although it appears more stable and less extreme. This suggests that the increase in AI generated contributions is not only caused by newly appearing repositories or sudden dataset expansion, but reflects a more genuine growth in AI usage over time.
Taken together, these results show that AI usage in collaborative software development is increasing, but temporal analyses of AI activity should be interpreted with caution. Without controlling for repository activity and dataset composition, such analyses may give an exaggerated view of how quickly or consistently AI usage is increasing.

6.3 AI Generated Pull Requests and Reviews In Practice

How AI generated pull requests are handled during the review process provides important insight into their role within collaborative software development. As shown in Table 29, a large portion of the recorded review activity on AI authored pull requests consists of comments, indicating that these pull requests are often discussed and examined rather than simply accepted without feedback. This pattern is also visible in Figures 4 and 5, which suggest that comments remain

the dominant review outcome for AI authored pull requests over time. It should be noted that detailed review outcome data for human authored pull requests was not available in the dataset, which limits direct comparison in this part of the analysis.

Another notable pattern observed in this study is the review behavior presented in Table 28, where AI authored reviews are primarily concentrated in the `commented` category. Together, these findings suggest that AI generated contributions are commonly incorporated into collaborative workflows through iterative discussion and reviewer feedback.

One possible interpretation is that AI generated pull requests often function as useful starting points rather than fully finalized contributions. While these pull requests may contain valid and relevant code changes, they may still require clarification, refinement, or alignment with repository specific expectations before they can be fully integrated. As a result, the review process appears to play an important role in shaping and validating AI generated contributions.

Overall, this indicates that AI systems are not only contributing code, but are also influencing the nature of collaboration by increasing the importance of review, discussion, and refinement within pull request based development workflows.

6.4 Task Type Differences Between AI and Human Contributions

The type of tasks for which AI generated contributions are primarily used provides further insight into the role of AI in software development. As shown in Table 30, AI authored pull requests are more frequently associated with development oriented task types such as `feat`, `fix`, and `docs`, which correspond to feature implementation, bug fixing, and documentation tasks. Human authored pull requests are also appear often in `feat` and `fix` categories, but additionally show a more balanced distribution across categories such as `chore`, `build`, and other maintenance related tasks.

This difference suggests that AI systems are more commonly applied to tasks that involve generating or modifying functional code, while human developers, although performing similar tasks, remain more engaged in activities related to project maintenance, configuration, and supporting infrastructure. These types of tasks often require a deeper understanding of repository specific context and long term project structure, which may be more difficult for current AI systems to fully capture.

One possible explanation is that feature development and bug fixing tasks are more directly aligned with the strengths of large language models, which are trained on large scale code corpora and have demonstrated strong performance in code generation from natural language specifications [27]. In contrast, maintenance oriented tasks may involve implicit knowledge about project organization and workflow, making them less suitable for automated generation.

Overall, these findings suggest that AI contributions are not evenly distributed across all types of development activities, but are instead concentrated in areas

where code generation plays a central role. This highlights a division of labor in which AI supports code production, while human developers retain a stronger role in maintaining and organizing the broader development environment.

6.5 Differences Across AI Coding Agents

Differences across AI coding agents show whether AI assisted development behaves similarly across tools or varies depending on the specific agent used. As shown in Figure 6, the review outcome distributions of the analyzed AI agents are broadly similar. Across all five agents, the majority of reviews fall into the `commented` category, followed by `approved`, while `changes_requested` and `dismissed` remain comparatively rare.

Although minor differences are visible, such as Devin showing a slightly higher approval share and GitHub Copilot showing a somewhat higher rate of change requests, these variations remain limited. Overall, reviewers appear to interact with pull requests generated by different AI agents in broadly similar ways.

A similar pattern can be observed in Figure 7, where all agents exhibit consistently high confidence scores in task classification, with only small differences in average values and standard deviation. This suggests that the pull requests produced by different agents are similarly clear and well defined in terms of their categorized development intent.

Taken together, these findings suggest that the differences between AI coding agents are relatively small. While some individual agents show slight variation, the overall collaboration patterns observed in this study remain largely similar across tools.

6.6 Limitations

Although this study provides useful insights into AI assisted development on GitHub, several limitations should be considered when interpreting the findings. First, the analysis is based on the AIDEV dataset, which represents a selected subset of GitHub repositories and AI authored pull requests. As a result, the findings may not fully generalize to all repositories, development environments, or forms of AI assisted software development.

Second, some parts of the analysis were constrained by the available data. In particular, detailed review outcome data for human authored pull requests was not available in the dataset, which limited direct comparison between AI and human pull request evaluation in certain parts of the study.

Third, the task type analysis relies on categorized pull request labels based on conventions such as Conventional Commits [11]. While this provides a useful way to compare development activities, these labels may not always perfectly capture the full intent or complexity of a contribution.

These limitations do not invalidate the findings, but they do suggest that the results should be interpreted within the scope of the dataset used in this study, rather than as a complete picture of AI activity in collaborative software development.

7 Conclusion

This study examined how AI generated contributions are integrated and evaluated within collaborative software development workflows on GitHub. Using the AIDEV dataset, the research analyzed repository level AI usage, temporal trends, review behavior, task type distributions, and differences across AI coding agents in order to better understand the role of AI in modern pull request based development.

The findings show that AI systems are already actively contributing to software projects, particularly through pull request creation, but their role in collaborative workflows remains largely supportive. AI generated contributions are common across many repositories and have increased over time, indicating a growing presence of AI assisted development in practice. At the same time, review authority and final evaluative decisions remain primarily human driven. AI authored pull requests are often handled through comments and discussion, while AI authored reviews are mainly concentrated in feedback oriented review behavior rather than approval or rejection.

The results also suggest that AI contributions are not evenly distributed across all development activities. While both AI and human authored pull requests are frequently associated with tasks such as feature implementation, bug fixing, and documentation, AI generated contributions appear to be more concentrated in these categories. Human contributors, show a more balanced distribution across a wider range of tasks including feature implementation, bug fixing, and documentation like AI generated contributions, but also maintenance and infrastructure related tasks.

In addition, only limited differences were observed across the analyzed AI coding agents, suggesting that their collaborative behavior is more similar than different within the context of this dataset.

Overall, the findings indicate that AI is becoming an increasingly visible part of collaborative software development, not as a replacement for human developers, but as a contributing participant within existing workflows. As AI coding agents continue to evolve and become more widely adopted, understanding how they are integrated into collaborative development environments will remain an important area for future software engineering research.

References

[1] S. Joksimovic, D. Ifenthaler, R. Marrone, M. De Laat, and G. Siemens, "Opportunities of artificial intelligence for supporting complex problem-solving: Findings from a scoping review," *Computers and Education: Artificial Intelligence*, vol. 4, p. 100138, 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2666920X23000176

[2] L. Banh, F. Holldack, and G. Strobel, "Copiloting the future: How generative ai transforms software engineering," *Information and Software Technology*, vol. 183, p. 107751, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0950584925000904

[3] P. Kokol, "The use of ai in software engineering: A synthetic knowledge synthesis of the recent research literature," *Information*, vol. 15, no. 6, 2024. [Online]. Available: https://www.mdpi.com/2078-2489/15/6/354

[4] H. Li, H. Zhang, and A. E. Hassan, "The rise of ai teammates in software engineering (se) 3.0: How autonomous coding agents are reshaping software engineering," 2025. [Online]. Available: https://arxiv.org/abs/2507.15003

[5] M. M. Rahman and C. K. Roy, "An insight into the pull requests of github," in *Proceedings of the 11th Working Conference on Mining Software Repositories*, ser. MSR 2014. New York, NY, USA: Association for Computing Machinery, 2014, p. 364–367. [Online]. Available: https://doi.org/10.1145/2597073.2597121

[6] G. Gousios, M. Pinzger, and A. v. Deursen, "An exploratory study of the pull-based software development model," in *Proceedings of the 36th International Conference on Software Engineering*, ser. ICSE 2014. New York, NY, USA: Association for Computing Machinery, 2014, p. 345–355. [Online]. Available: https://doi.org/10.1145/2568225.2568260

[7] S. Peng, E. Kalliamvakou, P. Cihon, and M. Demirer, "The impact of ai on developer productivity: Evidence from github copilot," 2023. [Online]. Available: https://arxiv.org/abs/2302.06590

[8] B. Yetiştiren, I. Özsoy, M. Ayerdem, and E. Tüzün, "Evaluating the code quality of ai-assisted code generation tools: An empirical study on github copilot, amazon codewhisperer, and chatgpt," 2023. [Online]. Available: https://arxiv.org/abs/2304.10778

[9] A. B. Shirin Pirouzkhah, Pavlína Wurzel Gonçalves, "The value of effective pull request description," 2026. [Online]. Available: https://arxiv.org/abs/2602.14611

[10] Q. Zeng, Y. Zhang, Z. Qiu, and H. Liu, "A first look at conventional commits classification," in *2025 IEEE/ACM 47th International Conference on Software Engineering (ICSE)*, 2025, pp. 2277–2289. [Online]. Available: https://doi.org/10.1109/ICSE55347.2025.00011

[11] Conventional Commits, "Conventional commits," 2025. [Online]. Available: https://www.conventionalcommits.org

[12] R. A. Poldrack, T. Lu, and G. Beguš, "Ai-assisted coding: Experiments with gpt-4," 2023. [Online]. Available: https://arxiv.org/abs/2304.13187

[13] GitHub, "Github copilot," 2025, accessed: 2026-03-26. [Online]. Available: https://github.com/features/copilot

[14] Cognition, "Devin," 2025, accessed: 2026-03-26. [Online]. Available: https://devin.ai

[15] Cursor, "Cursor," 2025, accessed: 2026-03-26. [Online]. Available: https://cursor.com

[16] OpenAI, "Openai codex," 2025, accessed: 2026-03-26. [Online]. Available: https://openai.com/index/openai-codex/

[17] Anthropic, "Claude code," 2025, accessed: 2026-03-26. [Online]. Available: https://claude.com/product/claude-code

[18] A. E. Hassan, G. A. Oliva, D. Lin, B. Chen, Z. Ming, and Jiang, "Towards ai-native software engineering (se 3.0): A vision and a challenge roadmap," 2026. [Online]. Available: https://arxiv.org/abs/2410.06107

[19] U. K. Durrani, M. Akpinar, M. F. Adak, A. T. Kabakus, M. M. Ozturk, and M. Saleh, "A decade of progress: A systematic literature review on the integration of ai in software engineering phases and activities (2013–2023)," *IEEE Access*, vol. 12, pp. 171 185–171 199, 2024.

[20] V. Terragni, A. Vella, P. Roop, and K. Blincoe, "The future of ai-driven software engineering," *ACM Transactions on Software Engineering and Methodology*, 2025.

[21] F. Song, A. Agarwal, and W. Wen, "The impact of generative ai on collaborative open-source software development: Evidence from github copilot," 2025. [Online]. Available: https://arxiv.org/abs/2410.02091

[22] Amazon Web Services, "Amazon codewhisperer," 2025, accessed: 2026-03-26. [Online]. Available: https://aws.amazon.com/codewhisperer/

[23] OpenAI, "Chatgpt," 2025, accessed: 2026-03-26. [Online]. Available: https://openai.com/chatgpt

[24] Hugging Face, "Hugging face datasets," 2025. [Online]. Available: https://huggingface.co

[25] W. McKinney *et al.*, "pandas: a foundational python library for data analysis and statistics," *Python for high performance and scientific computing*, vol. 14, no. 9, pp. 1–9, 2011.

[26] J. D. Hunter, "Matplotlib: A 2d graphics environment," *Computing in Science Engineering*, vol. 9, no. 3, pp. 90–95, 2007.

[27] M. Chen, J. Tworek, H. Jun, Q. Yuan, H. P. de Oliveira Pinto, J. Kaplan, H. Edwards, Y. Burda, N. Joseph, G. Brockman, A. Ray, R. Puri, G. Krueger, M. Petrov, H. Khlaaf, G. Sastry, P. Mishkin, B. Chan, S. Gray, N. Ryder, M. Pavlov, A. Power, L. Kaiser, M. Bavarian, C. Winter, P. Tillet, F. P. Such, D. Cummings, M. Plappert, F. Chantzis, E. Barnes, A. Herbert-Voss, W. H. Guss, A. Nichol, A. Paino, N. Tezak, J. Tang, I. Babuschkin, S. Balaji, S. Jain, W. Saunders, C. Hesse, A. N. Carr, J. Leike, J. Achiam, V. Misra, E. Morikawa, A. Radford, M. Knight, M. Brundage, M. Murati, K. Mayer, P. Welinder, B. McGrew, D. Amodei, S. McCandlish, I. Sutskever, and W. Zaremba, "Evaluating large language models trained on code," 2021. [Online]. Available: https://arxiv.org/abs/2107.03374